



Abstract

- Web-based platform to connect blood donors, recipients and blood banks.
- Allows users to register as donors and search for nearby blood banks.
- View available blood groups and send blood collection requests.
- Uses MySQL database, Node.js with Express.js for backend APIs, and React.js for a responsive user interface.
- Provides real-time location-based services for fast and efficient blood donation support.

Problem Statement & Objectives

Problem: Existing systems do not provide a centralized, real-time platform to find blood donors and blood banks based on location, leading to delays during emergencies.

Objectives:

- Develop a centralized platform for donor, recipient and blood bank registration.
- Enable location-based search for blood banks and donors.
- Allow users to send and manage blood collection requests.

Scope: Web application for donors, recipients, collectors and blood banks with real-time location and blood group search.

Existing vs Proposed System

Feature	Existing System	Proposed System
Donor Search	Manual, phone-based	Online search with filters
Data Storage	Registers / spreadsheets	Centralized MySQL database
Location Tracking	Not available	Real-time location-based
Coverage	Limited to single hospital	City-wide network
Blood Inventory	Not updated in real time	Live inventory by blood group
Response Time	Slow and delayed	Instant matching & requests

Feasibility Study

- **Technical:** Required skills, tools and platforms are readily available (React.js, Node.js, MySQL).
- **Economic:** Low cost using open-source stack and free-tier hosting.
- **Operational:** User-friendly interface with high user acceptance and easy maintenance.
- **Schedule:** Can be completed within the semester timeline.

Implementation Screenshots



Screenshot 1: Nearby blood banks page

Screenshot 2: Collector dashboard page

Results & Testing

- Successfully implemented user registration, login and role-based access.
- Blood bank search and donor search working with location.
- Collection request creation, accept/reject and status updates.
- All CRUD operations tested using Postman.
- Validations for form inputs and user authentication.
- The application responds quickly and works smoothly on web browsers.

Conclusion

- Developed a functional Blood Bank and Donor Locator App meeting all objectives.
- Provides a centralized, real-time platform for donors, recipients and blood banks.
- Helps reduce response time during blood emergencies.
- Efficient use of database systems and backend development concepts.

Future Scope

- Add mobile application (Android/iOS).
- Integrate real-time notifications (SMS/Email).
- Add AI-based donor matching.
- Support multi-city and nationwide deployment.
- Integrate with hospital management systems.
- Add analytics dashboard for blood demand prediction.

References

- [1] MDN Web Docs – React.js [2] Node.js Official Documentation [3] MySQL Documentation
[4] Google Maps API Documentation



GitHub Repository
(Scan QR)



Tools & Technologies



React.js
HTML, CSS, JS
(Front-End)



Node.js +
Express.js
(Back-End)
REST APIs



MySQL
(Database)

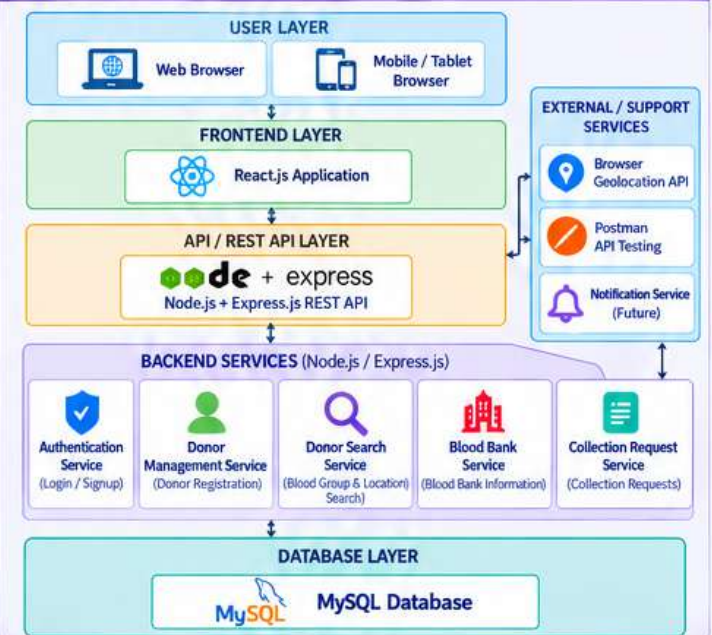


Google Maps API
(Location Services)



Git & GitHub
Postman
VS Code
Draw.io

System Architecture



ER Diagram & Database Design

